

# Influence of Road Proximity on the Concentrations of Heavy Metals in Korean Urban Agricultural Soils and Crops

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**Abstract** The urban agricultural (UA) environment near active roadways can be degraded by traffic-related particles (i.e., exhaust gases and road dust), which may contain heavy metals. The current study investigated changes in heavy-metal [cadmium (Cd), copper (Cu), chromium (Cr) nickel (Ni), lead (Pb) and zinc (Zn)] concentrations in soils located near highly trafficked roads in Korea and the subsequent uptake of these metals by Chinese cabbage. Heavy-metal plant concentrations were determined in both washed and unwashed plant leaves to determine whether foliar deposition played any role in plant metal uptake. Soil concentrations of Cd, Cu, Pb, and Zn were all lower than the Korean standard soil limits and showed no significant influence from road traffic. In contrast, both Ni and Cr concentrations in soils collected within 10 m of the road were 4 and 5 times greater, respectively, than those in soils collected 70 m from the road. Heavy-metal concentrations in unwashed Chinese cabbage leaf collected at 5 m from the road were consistently greater than those of washed leaf samples, thus indicating the deposition of traffic-related particles on the plant surface. With

the exception of Cu, all heavy-metal concentration in washed plant samples collected at 5 m also showed greater accumulation compared with samples collected further away. This was mainly attributed to increased total soil heavy-metal concentrations and increased metal phytoavailability induced by decreases in soil pH near the road. However, overall heavy-metal soil concentrations were well lower than the allowable concentrations, and the levels observed in plants collected in this study were considered not to currently pose a significant risk to human health. However, some traffic-related heavy metals, in particular Cr and Ni, were being accumulated in the roadside UA environment, which may warrant some caution regarding the environment and/or health issues in the future.

Urban agriculture (UA) today is commonly proposed as a sustainable solution for dealing with gaps in the local food-supply system because it has nutritional, economic, and social benefits (Armar-Klemesu 2000; Nugent 2000; Saldivar-Tanaka and Krasny 2004). UA can also increase a country's food security when defined as sufficient access by all people to enough food for a healthy and productive life in the face of continuing population growth (Pinstrup-Andersen 2009). Because Kulak et al. (2013) showed that land used on an urban fringe for community farming (26 ha) could potentially decrease greenhouse-gas emissions in Sutton, England, by  $\leq 881 \text{ t CO}_2\text{e a}^{-1}$ , UA can also mitigate global warming. For these reasons, UA is today regarded as a rapidly expanding and an increasingly popular practice worldwide.

However, although the potential benefits of UA are numerous, potential human health risks are associated with heavy-metal exposure because heavy metals are generally

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increased in many urban soils due to a variety of human activities including industrial activity, vehicle exhaust, waste disposal, and fuel combustion (Li et al. 2001; Wei and Yang 2010; Luo et al. 2012). For this reason, food crops produced in urban areas are more likely to be contaminated with heavy metals than those cropped in rural areas, and this is one of major concerns limiting UA adoption (Nabulo et al. 2012; Cruz et al. 2014; Kim et al. 2015b). For instance, McBride et al. (2014) reported that lead (Pb) concentrations in 13% of urban-garden soil samples collected from New York City and Buffalo, USA, exceeded the health-based guidance value ( $400 \text{ mg kg}^{-1}$ ). Subsequently, Pb concentrations in 47% of root crops and 9% of leafy greens exceeded the health-based guidance values (root crop =  $0.1 \text{ mg kg}^{-1}$ , leafy green =  $0.3 \text{ mg kg}^{-1}$ ).

Traffic activities are a major source of heavy metals in urban areas, and consequently much of the recent research interest has focused on traffic-related particles as one of the major urban environmental pollutants (Säumel et al. 2012; Wiseman et al. 2013). Heavy metal containing traffic-related particles is generated from tires, brakes, fuel, road surface, and general vehicle wear processes and can be transported into the soil and plant surface and eventually translocated to plant tissues (Hays et al. 2011; Yuen et al. 2012; Wiseman et al. 2013). Nabulo et al. (2006) reported that total cadmium (Cd), Pb, and zinc (Zn) concentrations in soils were highest in the soil directly adjacent to the trafficked road and decreased with distance  $\leq 30 \text{ m}$  from the road. Säumel et al. (2012) also showed that greater overall traffic burdens increased Pb concentration in crops so that 67% of the crops from sites with greater overall traffic exceeded the European Union standard for Pb (leafy vegetable =  $0.3 \text{ mg kg}^{-1}$ , fruit =  $0.1 \text{ mg kg}^{-1}$ , stem and root vegetable =  $0.1 \text{ mg kg}^{-1}$ ). Therefore, much more attention must be paid to the levels of heavy metals in traffic-derived particles because these particles are the main factor affecting the quality of urban agricultural product grown in roadside.

In 2011, when Korea legislated acts that supported and promoted UA, many urban communities became heavily involved in UA activities. According to Heo and Kwon (2014), approximately 770,000 urban dwellers now participate in urban farming on approximately 558 ha across Korea. However, despite the huge rate of UA practice, there is currently a lack of UA research concerned with the potential environmental risks associated with heavy-metal exposure from traffic activities. Therefore, the current study was performed to assess the influence of traffic-related particles on a roadside agricultural area in Namyangju, Korea, by determining the variation in concentrations of Cd, chromium (Cr), copper (Cu), nickel (Ni),

Pb, and zinc (Zn) in both roadside soils and plants with distance from the road.

## Materials and Methods

### Study Area

This study was performed in an UA field (0.8 ha) near a local four-lane road (road number = 383) located in Namyangju city, Korea (Fig. 1). This city was developed rapidly between the 1980s and early 2010s and now has a population of  $>500,000$ . Because this city is one of Seoul's satellite cities, bordering the northwest of Seoul (capital city), many people commute from this city to Seoul and vice versa. The average daily traffic (ADT) of the studied road was not directly determined here, but was determined from available secondary data, which—according to the Traffic Monitoring System (<http://www.road.re.kr>)—indicated that the ADT of the No. 383 local road located in another area was approximately 20,000 vehicles.

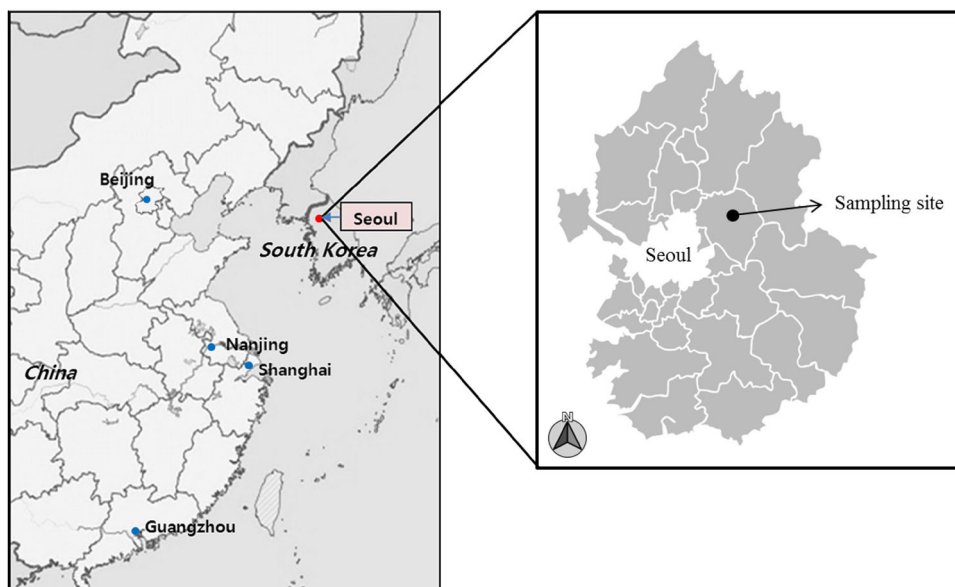
### Soil Samples

Upland soil samples were collected in October 2013 along a 50 m transect at 5 m intervals (Fig. 2). Soils between 60 and 70 m from the road (70 m hereafter) were also collected as reference soils because it was assumed that at this distance the soil would not be affected by the traffic activities of the road. A stainless-steel shovel was used to collect the top layer (0–20 cm) of the soil. From each transect, ten soil samples 5 m apart from each other were collected and then bulked to provide one composite sample (approximately 5 kg). Collected soils were air-dried, sieved ( $<2 \text{ mm}$ ), and stored in an acid-washed plastic container until further analysis.

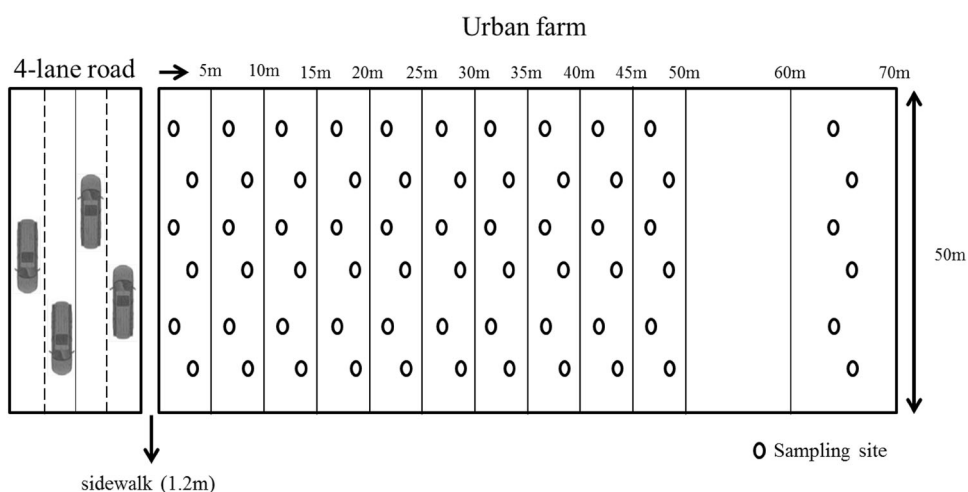
### Plant Samples

Chinese cabbage leaves were collected from the same location as the soil samples using pruning shears. A total of 30 leaf samples were collected from within each transect. On arrival in the laboratory, half of the collected plant leaf samples were first washed with tap water, and then with distilled water before being dried at  $60^\circ\text{C}$  for 48 h. The remaining unwashed leaf samples were dried at  $60^\circ\text{C}$  for 48 h without any washing procedure (Nabulo et al. 2010). This was performed to examine the contribution of deposited particles on the leaf surface to the increased metal concentration of the leaf samples. The dried plant samples were powdered using a commercial blender and used for all further analysis.

**Fig. 1** Location of urban agriculture field in Namyangju, Korea, from which soil and plant were sampled



**Fig. 2** Scheme of the urban agricultural area near road at different distances from the roadside edge



## Analysis of Soil and Plant

### Soil

Total heavy-metal (Cd, Cr, Cu, Ni, Pb, and Zn) soil concentrations were determined using inductively coupled plasma-optical emission spectroscopy (ICP-OES) according to an aqua regia digestion method. Specifically, soil (3 g) was digested with aqua regia (28 mL) using a commercial trace metal digestion system (SMA 20A, Gerhardt, UK), and the suspension was subsequently diluted to 100 mL with 0.5 M HNO<sub>3</sub>. According to this procedure, samples were filtered through Whatman no. 42 filter papers before determination of total metal concentrations using ICP-OES (iCAP 6000 series, Thermo Scientific, UK). Phytoavailable soil metal concentrations were determined using 1 M NH<sub>4</sub>NO<sub>3</sub> extractions (DIN 1995). Briefly, soil

(10 g) was extracted with 1 M NH<sub>4</sub>NO<sub>3</sub> (25 mL) with shaking on an end-over-end shaker for 2 h, and the extracted solution were filtered through 0.45 μm filters followed by determination of metal concentrations in the solution by ICP-OES. Because pH is one of the main soil properties influencing phytoavailability (Kim et al. 2015a), to investigate the relationships between phytoavailable metal concentrations and soil pH, pH was measured on a 1:5 soil-to-distilled water suspension using a pH meter (MP220, Mettler Toledo, UK) after 1 h of shaking. All of the soil samples were analyzed in triplicate and the results averaged.

### Plant

Powdered sample (0.5 g) was digested in triplicate with 10 mL of nitric acid using a block digester (DS20, Foss

Tecator, Sweden) followed by dilution to 50 mL using distilled water. The solution was filtered through Whatman (no. 42) filter papers, and ICP-OES was used to measure heavy-metal (Cd, Cr, Cu, Ni, Pb, and Zn) concentrations in the solutions.

To ensure the quality of the analysis process, the same procedure was applied to certified reference materials (CRMs) including BAM-U112 for soil and spinach leaves (SRM 1570a) for plant analysis. The recovery for Cd, Cr, Cu, Ni, Pb, and Zn in BAM-U112 ranged from 88–103%, 89–111%, 87–108%, 91–108%, 89–109%, and 92–108%, respectively, and those for Cd, Cu, Ni, and Zn in the spinach leaves were in the range of 91–106%, 89–110%, 92–116%, and 87–112%, respectively.

### Data Treatment and Statistics

For ease of visualization, all data listed in tables and figures were mean values with errors corresponding to one SD of the mean. Any significant differences amongst plant samples were determined by analysis of variance using SAS 9.3 software (SAS for Windows v. 9.3, SAS Institute Inc., Cary, North Carolina, USA).

## Results and Discussion

### Total Heavy-Metal Concentrations in Soils

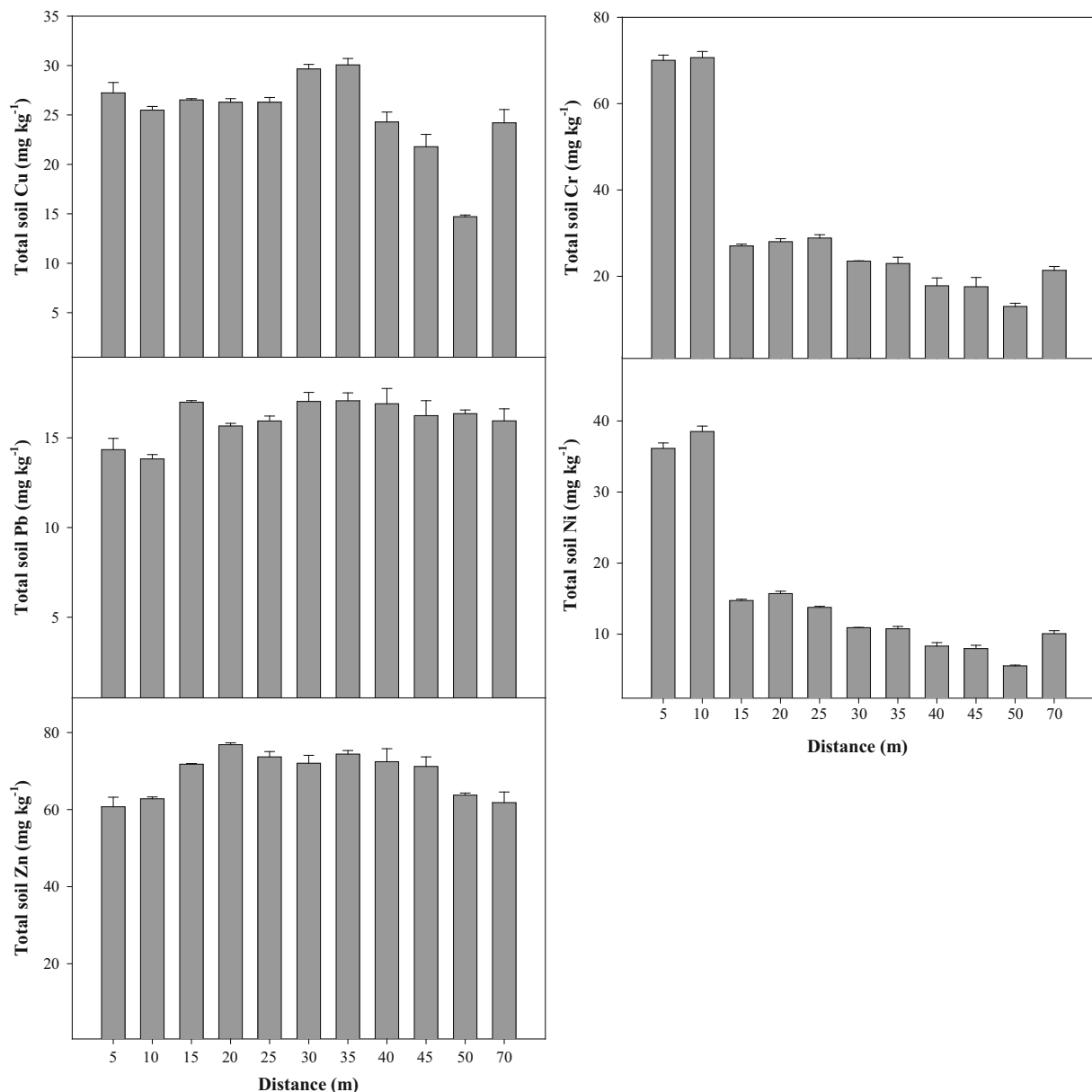
Regardless of the distance from the road, heavy-metal soil concentrations were consistently lower than the Korean standard guideline values for an agricultural soil (MOE 2010), and Cd was not detected at all (Fig. 3). The concentration of soil Cu ranged from 15 to 30 mg kg<sup>-1</sup>, which was similar to the levels typically found in normal cultivated upland soil (Cu = 23 mg kg<sup>-1</sup>) throughout Korea (MOE 2014). Pb and Zn concentrations varied from 14 to 18 mg kg<sup>-1</sup> and from 61 to 77 mg kg<sup>-1</sup>, respectively, which were also both lower than the typical levels experienced in normal cultivated upland soil (Pb = 22 mg kg<sup>-1</sup>, Zn = 82 mg kg<sup>-1</sup>) throughout Korea (MOE 2014). Although there was no dramatic variation in Cu, Pb, and Zn concentrations with distance from the road, indicating no significant affect from traffic activities at this site, there were some slight variations. For Pb and Zn, soils adjacent to the road (<10 m) showed slightly lower concentrations than those measured in soils >15 m from the road.

In contrast, much more significant variations in Cr and Ni soil concentrations were observed with distance from the road. Within 10 m (Cr = 70–71 mg kg<sup>-1</sup>, Ni = 36–39 mg kg<sup>-1</sup>) were 2.8- and 2.1-fold greater, respectively, than the expected background levels for the corresponding heavy metal. The soils adjacent to the road (<10 m) also had far greater Cr and

Ni concentrations than those measured in soils >15 m from the road. The Cr and Ni soil concentrations beyond 15 m ranged from 13 to 33 mg kg<sup>-1</sup> and from 6 to 16 mg kg<sup>-1</sup>, respectively, which was generally lower than natural background levels typically found for Cr (25 mg kg<sup>-1</sup>) and Ni (18 mg kg<sup>-1</sup>) throughout Korea (Yoon et al. 2009). This implied that although there was a significant contribution from road activities, which increased both Cr and Ni soil concentrations, the area affected was limited to soils adjacent to the road.

The concentrations of Cu, Pb, and Zn in soils adjacent to the road were expected to be greater than those in the soils collected further away because previous studies had shown that the highest concentration of these metals were generally observed in soils collected nearest to road where distribution was influenced by transported pollutants by way of airflow, runoff water, and splashing water from road (Werkenthin et al. 2014). For example, Swaileh et al. (2004) previously showed that Cu, Pb, and Zn concentrations in soils collected at the roadside edge were approximately 25, 42, and 7 times greater than those of soils collected at a 200 m distance, respectively. This increase in roadside metal concentration was attributed to traffic-related metals derived from combustion of leaded fuel, lubricating motor oil, vehicle tire wear, and brake pads (Swaileh et al. 2004). Other studies have also confirmed that the concentrations of heavy metals (Cd, Cr, Cu, Ni, Pb, and Zn) in soil along the oldest federal highway (the AVUS Autobahn) in the world were ≤20 times greater than a reference site 800 m from the roadside edge (Kluge and Wessolek 2012). Many studies have indicated that the major contributor to increased Pb levels in roadside soil was leaded gasoline. However, in this study, increased levels of Cu, Pb, and Zn close to the roadside were not observed, thus implying that proximity to the roadside was not a significant influence on the levels of these metals in soils. Even if in the case of Cu and Zn the reason for this was not explainable with our data, the observations are easier to explain for Pb. Most older studies were performed when leaded gasoline use was common. However, today because leaded gasoline was completely replaced with unleaded gasoline throughout Korea in 1993, it can be assumed that Pb inputs from road traffic are no longer a significant factor.

In direct contrast to Cu, Pb, and Zn, the Cr and Ni concentrations in soil near the road (<10 m) were 4 and 5 times greater, respectively than those in soil collected from the reference site (70 m) (Fig. 3). Furthermore, total Cr and Ni concentrations in the soils were positively correlated ( $R^2 = 0.99$ ;  $p < 0.0001$ ), thus implying that Cr and Ni were released concomitantly from the same source. This is consistent with efflux by way of metal alloy wear particles emitted from vehicle engines because studies of



**Fig. 3** Total heavy-metal concentrations in roadside soils at different distances from the roadside edge

correlations amongst metals [arsenic (As), beryllium, Cd, cobalt (Co) Cr, manganese, Ni, Pb, antimony, and selenium] in PM<sub>10</sub> (particulate matter with an aerodynamic diameter <10 µm) collected near a highway found a significantly strong relationship between Ni and Cr (Hays et al. 2011). However, Cr and Ni are also found in brake- and tire-wear debris in addition to engine-wear particles (Yuen et al. 2012). Although a wide range of metals—including As, Cd, Cu, Pb, and Zn—can find their way into agricultural soils, independently of vehicle traffic, through application of agricultural related materials such as compost, chemical fertilizers, and

pesticides (Huang et al. 2007; Atafar et al. 2010; Kelepertzis 2014), these pathways are very unlikely to carry Ni and Cr to soil; thus, their presence in soils must be attributed to road activity. In some cases, the addition of municipal solid waste compostor tannery sludge compost has increased total Cr concentrations in soil (Bhattacharyya et al. 2005; Silva et al. 2010); however, such materials are not legally used for the manufacture of agricultural composts in Korea. Therefore, it was concluded that the increased Cr and Ni soil concentrations observed in this study were unquestionably derived from vehicle exhausts.

## Phytoavailable Heavy-Metal Concentrations and Soil pH

Total heavy-metal concentrations by themselves are often a poor indicator of crop exposure. A much better estimate of metal crop exposure is gained by considering the phytoavailable metal pool present in the soil (Menzies et al. 2007). In this study, metal phytoavailability was determined using 1 M  $\text{NH}_4\text{NO}_3$  extraction (DIN 1995). However, only Ni phytoavailability (Fig. 4) is discussed in detail here because all of the other heavy metals studied were lower than the detection limit of the 1 M  $\text{NH}_4\text{NO}_3$  extraction method due to their different adsorption ability. The affinity order of the metal cations onto kaolinite and montmorillonite variable charged sites were  $\text{Pb} > \text{Cu} > \text{Zn} > \text{Cd} \approx \text{Ni}$  and  $\text{Pb} \gg \text{Cu} > \text{Zn} > \text{Cd} > \text{Ni}$ , respectively (Gu and Evans 2007, 2008). Therefore, Ni is considered more mobile and phytoavailable in soil than other metals.

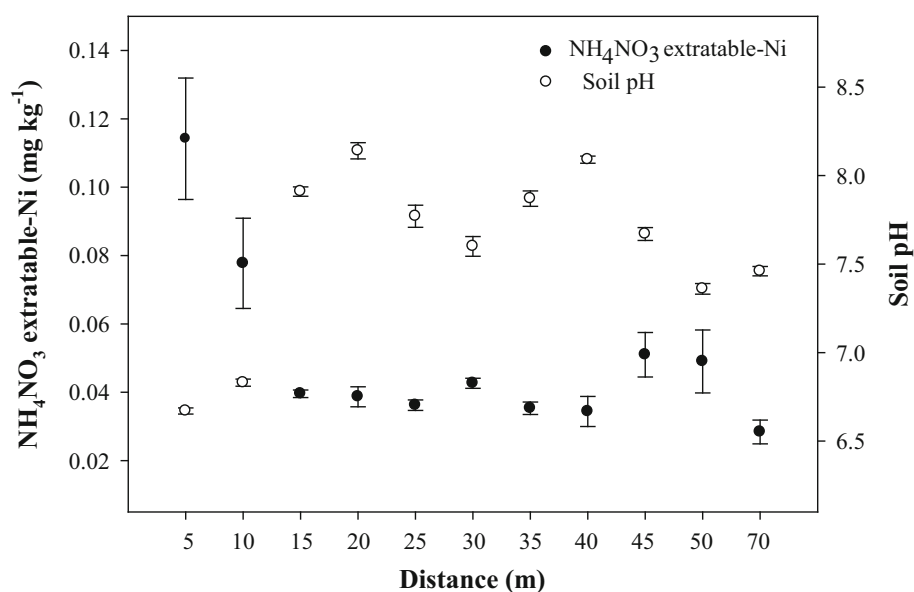
Phytoavailable Ni concentrations in soil—being 4.1 times greater at 5 m than that of the reference site at 70 m—were greater in soils adjacent to the road than in soils further away. Greater Ni phytoavailability was attributed simply to greater total Ni concentrations (Fig. 3) and lower soil pH (Fig. 4) in soils closer to the road. Soil pH in samples collected at 5 and 10 m were approximately 1 unit lower than those collected between 15 and 70 m (Fig. 4). Soil pH is one of the most important factors affecting heavy-metal phytoavailability because phytoavailability increases with decreasing soil pH (Bolan et al. 2014; Kim et al. 2015a). In general, urban soils tend to have alkaline pHs due to the presence of alkalizing products such as cement, concrete, mortar, and atmospheric-particulate deposition (De Kimpe and Morel 2000;

Li et al. 2013). However, the pH of the soils collected at 5 m and 10 m were 6.7 and 6.8, whereas the pH of the soils collected between 15 and 70 m ranged from 7.4 to 8.1, which was similar to the typical pH of an urban soil. The decreased soil pH closer to the road was assumed to be a contribution of reactive-nitrogen deposition. Most of the reactive N from vehicular exhaust is deposited within 10 m of roads, and this reactive N can acidify roadside soils (<10 m) due to nitrification of the deposited N (Rossi et al. 2015). In agreement with our observations, Stevens et al. (2009) showed a decrease in soil pH and increases in KCl-extractable soil Pb concentration following the deposition of reactive nitrogen and sulfur. In close agreement with this previous study, the deposition of reactive N sourced from vehicular exhaust to the roadside soil decreased soil pH and subsequently increased phytoavailable Ni pools ( $R = 0.91$ ,  $p = 0.001$ ). Thus, increased Ni phytoavailability near the road was attributed to a combination of greater total Ni concentration and lower pH, which could potentially enhance accumulation of Ni by Chinese cabbage when grown in the study site.

## Heavy Metals in Plant Tissues

To quantify the contribution of air deposition on plant surfaces to the final heavy-metal concentrations in cropped plants, metal concentrations in unwashed and washed plant samples were compared (Table 1). For all metals considered, unwashed samples consistently had greater total metal concentrations than washed samples, where the difference was greater for plants collected within 10 m of the road. This indicated that there was a significant influence of road traffic on increasing some phytoavailable heavy-metal

**Fig. 4**  $\text{NH}_4\text{NO}_3$  (1 M)-extractable Ni concentrations in roadside soils and soil pH at different distances from the roadside edge



**Table 1** Concentration of heavy metals in plants ( $\pm$ SD) grown in roadside soils at different distances from the roadside edge [means in each column followed by a different small letter are significantly different ( $p < 0.05$ )]

| Distance (m) | Cd ( $\text{mg kg}^{-1}$ ) |          | Cr ( $\text{mg kg}^{-1}$ ) |            | Cu ( $\text{mg kg}^{-1}$ ) |            | Ni ( $\text{mg kg}^{-1}$ ) |            | Pb ( $\text{mg kg}^{-1}$ ) |        | Zn ( $\text{mg kg}^{-1}$ ) |           |
|--------------|----------------------------|----------|----------------------------|------------|----------------------------|------------|----------------------------|------------|----------------------------|--------|----------------------------|-----------|
|              | Unwashed                   | Washed   | Unwashed                   | Washed     | Unwashed                   | Washed     | Unwashed                   | Washed     | Unwashed                   | Washed | Unwashed                   | Washed    |
| 5            | 0.52a, A                   | 0.51a, A | 8.86a, A                   | 4.39a, B   | 27.5ef, A                  | 24.8 cd, B | 5.48a, A                   | 3.23a, B   | 1.47a, A                   | 0.29B  | 110.0a, A                  | 106.2a, B |
| 10           | 0.31b, A                   | 0.22b, B | 5.82b, A                   | 3.85b, B   | 28.6ef, A                  | 24.8 cd, B | 4.38b, A                   | 2.67b, B   | 0.68b                      | ND     | 51.1b, A                   | 47.8b, B  |
| 15           | ND                         | ND       | 3.43c, A                   | 3.08c, B   | 32.5c, A                   | 30.5b, A   | 2.10c, A                   | 1.90def, B | 0.24c                      | ND     | 28.9ef, A                  | 20.7f, B  |
| 20           | ND                         | ND       | 3.49c, A                   | 3.10c, B   | 35.3b, A                   | 32.7b, A   | 2.18c, A                   | 2.11c, A   | ND                         | ND     | 31.1de, A                  | 27.6d, B  |
| 25           | ND                         | ND       | 3.38c, A                   | 3.02 cd, B | 30.0de, A                  | 25.9 cd, A | 2.09c, A                   | 1.98cde, A | ND                         | ND     | 24.2 g, A                  | 23.8e, B  |
| 30           | ND                         | ND       | 3.31c, A                   | 3.20c, A   | 39.5a, A                   | 35.9a, A   | 2.08c, A                   | 2.05 cd, A | ND                         | ND     | 23.4 g, A                  | 21.6ef, B |
| 35           | ND                         | ND       | 2.87d, A                   | 2.56f, B   | 31.8 cd, A                 | 31.0b, A   | 1.86d, A                   | 1.69 g, A  | ND                         | ND     | 23.0 g, A                  | 21.7ef, B |
| 40           | ND                         | ND       | 2.82d, A                   | 2.65ef, B  | 34.0bc, A                  | 31.1b, A   | 1.78d, A                   | 1.77 fg, A | ND                         | ND     | 27.2f, A                   | 24.2e, B  |
| 45           | ND                         | ND       | 3.41c, A                   | 2.83de, B  | 27.1f, A                   | 26.7c, A   | 2.19c, A                   | 1.98cde, A | ND                         | ND     | 37.3c, A                   | 32.4c, A  |
| 50           | ND                         | ND       | 2.79d, A                   | 2.75ef, A  | 33.0bc, A                  | 29.9b, A   | 1.83d, A                   | 1.83efg, A | ND                         | ND     | 29.4ef, A                  | 29.6d, A  |
| 70           | ND                         | ND       | 2.81d, A                   | 2.61 efA   | 23.4 g, A                  | 22.8d, A   | 1.82d, A                   | 1.74 fg, A | ND                         | ND     | 30.3d, A                   | 28.6d, A  |

Significant differences between washed and unwashed plant samples are denoted by different capital letters ( $p < 0.05$ )n.d not detected ( $<0.001 \text{ mg kg}^{-1}$ )

concentrations in the urban agricultural environment adjacent to the road. Increased heavy-metal contributions to the roadside environment could occur by way of splash or water runoff from the road and/or the transport of airborne traffic-related particles. Werkenthin et al. (2014) reported that soil collected within 5 m of a road was influenced by runoff water from the road and that soil collected within 10 m was also influenced by splash water. Data for unwashed samples showed that airborne deposition occurred  $\leq 45$  m from the road because Cr concentrations in unwashed plant samples collected 0–45 m from road were generally greater than those of the corresponding washed samples. Again, it seems that Cr and Ni were released from the road concomitantly because the amount of Cr deposited on the plant surface (the difference between Cr concentration in unwashed plant and Cr concentration in washed plant) correlated positively with the amount of Ni deposited on the plant surface ( $R = 0.96$ ,  $p < 0.0001$ ). Previous studies have reported that traffic-related particles deposited on plant surfaces increased the human health risk when ingested without washing (Nabulo et al. 2006; Kim et al. 2015b). In line with this, our data also suggest that vegetables grown in roadside soils ( $< 50$  m) should be thoroughly washed before ingestion to minimize any potential health risk.

Heavy-metal concentration determined in washed plant samples were assumed to have been accumulated in the plant tissues through direct uptake by plant roots from soil. Amongst all washed samples, leaves of Chinese cabbage collected at 5 m showed the highest concentration of all heavy metals except for Cu. Greater Cr and Ni concentration in tissues of plants situated near the road were expected due to the increased concentration of these metals in the soils and especially due to the greater Ni phytoavailability induced by a decrease in soil pH near the road. Ni concentrations in washed plant samples were positively correlated with phytoavailable Ni pools in soils ( $R = 0.96$ ,  $p < 0.0001$ ).

Interestingly, Cd, Pb, and Zn concentrations in plant tissue nearer the road also appeared to be greater than those collected further away from the road even though no significant difference in the total metal concentrations in the soils with distance from the road was observed. Cd concentrations in washed plant samples collected at 5 and 10 m from the road edge were 0.22 and 0.51  $\text{mg kg}^{-1}$ , respectively, whereas Cd was lower than the detection limit at distances  $> 10$  m. Likewise, Pb (0.29  $\text{mg kg}^{-1}$ ) was only detected at 5 m. Zn in washed plant samples collected close to the road (5 and 10 m) also showed a much greater accumulation of Zn compared with samples collected further away. Although the 1 M  $\text{NH}_4\text{NO}_3$ -extractable (phytoavailable) fractions were lower than detection limits, the enhanced accumulation of all of these

metals by Chinese cabbage near the road was probably due to the increased phytoavailability induced by decreases in soil pH. This interpretation is supported by Stevens et al. (2009), who showed that deposition of reactive N contributed to increases in the mobile pools of heavy metals, such as Pb, through acidification (Stevens et al. 2009; Rossi et al. 2015).

Unlike the other heavy metals studied, Cu concentrations in washed plant samples collected within 10 m of the road were relatively lower than those samples collected further away. This was probably due to a decrease in soil dissolved organic carbon (DOC) induced by a decrease in soil pH. Phytoavailability of Cu in soil is primarily governed by DOC concentration because it is increased through the formation of dissolved Cu–DOC complex as DOC concentration is increased (Ahmad et al. 2012; Kim et al. 2012). In contrast, a decrease in soil pH is likely to trigger precipitation of DOC onto soil particles. A decrease in soil pH leads to protonation of the functional groups resulting in decreasing net negative charge of the surface and, eventually, precipitation on soil particles (You et al. 1999). Kim et al. (2012) showed that in general, DOC concentrations in soil were positively correlated with soil pH. Thus a decrease in soil DOC near the road would occur with the observed soil pH decrease; subsequently, any Cu complexed with DOC would also be sorbed to soil particles.

Overall, the concentration of metals accumulated in Chinese cabbage grown nearby a road would not be a significant health risk to humans because such concentrations were much lower than the normal levels typically observed in this plant (Choi et al. 2010). Nevertheless, it is worth emphasizing that the proximity of a highly trafficked road may significantly effuse traffic-related particles into the local environment, thus resulting in increased heavy-metal deposition and changes in soil pH, which subsequently affects the magnitude of heavy-metal accumulation in plant tissue.

## Conclusion

The current study showed that the proximity of a road significantly influenced the roadside agricultural environment. As evidenced from the increased soil concentrations of Cr and Ni, as well as the increased levels of all metals in Chinese cabbage near the road, heavy metals commonly associated with road dust particles entered the agricultural environment. In addition, soil pH was significantly decreased near the road. Thus, Chinese cabbage in close proximity to the road accumulated more heavy metals than those grown further away. This was attributed to a combination of (1) increased total metal concentrations in soils

and (2) increased phytoavailability of the corresponding metals induced by a decrease in soil pH. Despite the observed increases in heavy-metal loadings, there was no immediate health concern with the metal concentrations in Chinese cabbage because they were consistently less than permissible levels. However an important outcome of this study is the demonstration that the UA environment near a road is potentially affected by traffic-related particles and thus requires further in-depth study.

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